

Addiction as Extractive Oscillator with Sensor Degradation: A Substrate-Independent Formal Theory from Chemical Dependence to Neural Interface Addiction

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Abstract

We develop a rigorous, substrate-independent mathematical framework for addiction. We demonstrate that addictive dynamics—whether chemical, behavioral, or neurostimulatory—constitute instantiations of a general dynamical systems class: the *extractive oscillator with sensor degradation* (EOS). The binge–remission–tension–relapse cycle, the progressive erosion of self-assessment capacity, the sunk-cost entrenchment mechanism, and the social isolation of addicted individuals are shown to be necessary structural consequences of a coupled two-agent system satisfying five axiomatic conditions. The framework is extended in three directions. First, we incorporate the neurobiochemical substrate, modeling the developmental decline of endogenous reward hormones as a time-dependent vulnerability parameter. Second, we introduce a novel threat analysis of neural interface addiction—direct electrical or optogenetic stimulation of reward centers via implanted devices—showing that this represents the limiting case where the effort-to-reward pathway length approaches zero, maximizing system instability. Third, and most extensively, we develop a formal theory of intervention, in which each therapeutic strategy is derived as a structural attack on a specific theorem, feedback loop, or phase transition of the model. The theory of positive alternatives to addiction is thereby transformed from a set of clinical recommendations into a mathematically grounded engineering discipline. The framework generates falsifiable predictions that distinguish it from purely descriptive clinical accounts and applies with equal force to heroin, social media, gambling, and future neural implant technologies.

We situate the EOS framework within the broader dynamical systems theory (DST) of psychiatric disorders (Kriger, 2026; Scheffer et al., 2024a,b), showing that the addiction limit cycle is a specific instantiation of the disorder attractor, the sensor degradation mechanism is the addiction-specific form of anosognosia, and the sunk-cost entrenchment is the cognitive mechanism underlying the progressive deepening

of the potential well. A formal correspondence table maps all EOS variables to the DST state-space representation, establishing that the two frameworks are mutually consistent and jointly generate predictions that neither produces alone.

Keywords: addiction, dynamical systems, extractive oscillator, sensor degradation, neural interfaces, reward hijacking, substrate independence, intervention theory, positive alternatives, relaxation oscillator, sunk-cost trap, adolescent vulnerability, attractor landscape, anosognosia

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Contents

1	Introduction	5
1.1	The Problem: Phenomenology Without Structure	5
1.2	The Approach: Addiction as an Extractive Oscillator System	5
1.3	Substrate Independence and the Neural Interface Horizon	5
1.4	The Intervention Problem	6
1.5	Relationship to the Dynamical Systems Theory of Psychiatric Disorders . .	6
1.6	Empirical Foundation	7
1.7	Plan of the Paper	7
2	The Agent Structure of Addiction	7
2.1	Reinterpretation of the Two-Agent Framework	7
2.2	Variable Mapping	8
2.3	Verification of the Five EOS Conditions	8
3	The Reward System as Exploited Substrate	9
3.1	The Natural Reward Circuitry	9
3.2	The Hijacking Mechanism	9
3.3	The Inverse Proportionality Principle	9
3.4	The Olds–Milner Theorem	9
4	The Addiction Cycle as Relaxation Oscillator	10
4.1	State Variables and Governing Equations	10
4.2	Hysteretic Switching and the Limit Cycle	10
5	Degradation of Self-Assessment Capacity	11
5.1	The Epistemic Model	11
5.2	Relationship to the DST Anosognosia Model	11
5.3	Empirical Confirmation	12
6	The Sunk-Cost Entrenchment Mechanism	12
7	Social Environment as Informational Isolation	12
7.1	The Star Network and Its Disruption	12
7.2	The One-Addict Principle	13
7.3	The Contagion Dynamics	13
7.4	Correspondence with DST Social Isolation Dynamics	13

8	The Developmental Vulnerability Window	13
8.1	The Endorphin Decline Model	13
8.2	The Vulnerability Window	14
9	Neural Interface Addiction: The Limiting Case	14
9.1	The Technology	14
9.2	The Zero-Path-Length Limit	14
9.3	Predictions for Neural Interface Addiction	14
9.4	The Black Market Scenario	15
10	Formal Correspondence with the DST Framework	16
11	A Formal Theory of Intervention	17
11.1	From Clinical Intuition to Structural Engineering	17
11.2	Loop 1: The Investment–Rationalization Loop	17
11.3	Loop 2: The Precision Erosion–Acceptance Loop	18
11.4	Loop 3: The Isolation–Dependence Loop	18
11.5	Combined Interventions: The Three-Loop Attack	19
11.6	The Vulnerability Window Interventions	19
11.7	The “Just for Today” Principle: A Formal Interpretation	19
11.8	Methadone and Substitution Therapy: A Formal Interpretation	19
11.9	Intervention Design for Neural Interface Addiction	20
11.10	Summary: The Intervention Taxonomy	20
12	Falsifiable Predictions	21
13	Discussion and Conclusion	22
13.1	Summary	22
13.2	Empirical Support from the Diversity of Addictions	22
13.3	The Deepest Structural Insight	24
13.4	Integration with the DST Framework	24
13.5	The Urgency of the Neural Interface Question	24
13.6	The Intervention as Engineering	24
13.7	Ethical Note	25
13.8	Limitations and Future Directions	25
	Appendices	28
A	Full Proof of the Limit Cycle Existence	28
A.1	The Planar System	28
A.2	Construction of the Trapping Region	28
A.3	Period Derivation	29
B	Full Proof of the Phase Transition in Self-Assessment	29
B.1	Setup and Precision-Weighted Updating	30
B.2	Precision Dynamics and the Phase Transition	30
C	Numerical Worked Example: Opioid Relapse Dynamics	31
C.1	Parameter Estimation from Empirical Data	31
C.2	Numerical Simulation	32

D	Relationship to Competing Models of Addiction	32
D.1	Redish (2004): Addiction as Computational Process Gone Awry	32
D.2	Koob & Le Moal (2001): Allostatic Model	33
D.3	Robinson & Berridge (2003): Incentive-Salience Theory	33
D.4	Borsboom (2017): Network Theory of Mental Disorders	33
D.5	Gutkin et al. (2006): Dynamical Systems Model of Nicotine Addiction . . .	34
D.6	Summary: What EOS Uniquely Explains	34
E	DST Mathematics for Substance Use Disorders	34
E.1	What DST Adds to the EOS Framework: Five Concrete Contributions . .	34
E.2	Core Model: Stochastic Dynamics on a Potential Landscape	36
E.3	Resilience and Kramers Escape Rate	36
E.4	Critical Slowing Down	36
E.5	The Addiction Potential (DST Appendix G)	36
E.6	Tolerance as Allostatic Shift	36
E.7	Hysteresis and Scarring	37
E.8	DSM-5-TR Severity Mapping	37
E.9	Anosognosia as Distance-Dependent Insight Decay	37
E.10	Social Connectivity as Stabilizing Feedback	37
E.11	Quasipotential for Oscillatory Dynamics	37
F	The Kriger System: General Definition	38
F.1	Verified Instantiations	39

1 Introduction

1.1 The Problem: Phenomenology Without Structure

The clinical literature on addiction is vast and sophisticated. Neurobiological models describe the dopamine system, the role of the prefrontal cortex, and the mechanisms of tolerance and withdrawal (Volkow et al., 2016; Koob & Le Moal, 2001). Psychological models address motivation, coping, trauma, and the stages of change. Sociological models examine the influence of environment, culture, and policy. Yet these accounts remain largely descriptive: they catalogue *what* happens without deriving it from structural first principles. The field lacks a unified formal framework from which the observed phenomena—the oscillatory cycle, the progressive self-deception, the entrenchment, the social isolation—can be deduced as necessary consequences of a small set of axioms.

The same gap afflicts intervention design. Current therapeutic approaches—cognitive-behavioral therapy, motivational interviewing, 12-step programs, harm reduction, pharmacotherapy—are grounded in clinical experience and empirical evidence of varying quality, but lack a formal basis that would explain *why* they work when they work, *why* they fail when they fail, and *how* they should be combined. Without a structural model, intervention design remains an art rather than an engineering discipline.

1.2 The Approach: Addiction as an Extractive Oscillator System

The present work addresses both gaps simultaneously. We identify addiction as a specific instantiation of a recently formalized class of dynamical systems: the *extractive oscillator with sensor degradation* (EOS). In Kriger (2017), the author introduced the EOS framework: a coupled two-agent dynamical system satisfying five structural conditions—asymmetric extraction (K1), hysteretic oscillation (K2), sensor degradation (K3), sunk-cost trapping (K4), and informational isolation (K5). That work demonstrated that these five conditions generate identical mathematical structures across radically different substrates: interpersonal relationships, political regimes, organizational dynamics, algorithmic platforms, and exploitative professional arrangements.

Here we develop the *addictive instantiation* in full detail. The addictive substance or behavior plays the role of the extracting agent; the addicted individual plays the role of the resource agent. This identification is not metaphorical. The governing equations, existence theorems, and structural predictions are mathematically identical to those of the general framework. Only the identity of the variables changes.

1.3 Substrate Independence and the Neural Interface Horizon

A formal theory of addiction tied to a specific neurochemical mechanism risks obsolescence. The development of neural interface technologies raises the concrete possibility of direct electrical or optogenetic stimulation of reward centers, bypassing neurochemistry entirely. Such technologies could create addictions with no chemical substrate: no molecule, no receptor, no pharmacokinetic profile—only a current through an electrode into the nucleus accumbens.

The framework developed here is substrate-independent by construction: the five EOS conditions make no reference to chemistry, pharmacology, or any specific physical mechanism. They describe the abstract structure of extraction, oscillation, sensor degradation,

entrenchment, and isolation. This structure is satisfied equally by heroin, by Instagram, by a slot machine, and by a hypothetical neural implant.

1.4 The Intervention Problem

If addiction has a formal structure, then intervention can be formalized as well. The central insight of this paper is that each component of the EOS framework—each theorem, each feedback loop, each phase transition—identifies a specific point of structural vulnerability. An intervention that targets a specific vulnerability is not a clinical recommendation; it is a *structural attack*. This transforms intervention design from intuition-guided practice to a discipline in which every strategy has a formal justification, a predicted mechanism of action, a predicted failure mode, and a measurable criterion of effectiveness.

Section 11 develops this theory of structurally targeted intervention in detail. We show that the full repertoire of known interventions—from the Narcotics Anonymous principle of “just for today” to methadone maintenance to environmental relocation—can be derived as special cases of a small number of structural operations on the EOS system. We also identify interventions that are predicted by the framework but have not yet been systematically implemented.

1.5 Relationship to the Dynamical Systems Theory of Psychiatric Disorders

The present work is situated within, and extends, the unified dynamical systems theory (DST) of psychiatric disorders developed in Kriger (2026), which itself builds upon the conceptual framework of Scheffer et al. (2024a,b). The DST framework models the healthy mind as a stable attractor on a potential landscape, psychiatric disorders as alternative attractors, and clinical deterioration as a critical transition governed by tipping points, hysteresis, and critical slowing down. Substance use disorders are treated in that work as one of nine diagnostic categories (Appendix G of Kriger, 2026), modeled via a progressive deepening of the disorder attractor through neuroadaptation.

The present article provides the *internal mechanics* of the addiction attractor that the DST framework describes topologically. Specifically:

- (i) The DST framework represents addiction as a point attractor with increasing depth $\Lambda_{\text{add}}(H)$. We show that the internal dynamics are better described as a *limit cycle* (the binge–remission–tension–relapse oscillation), consistent with the DST treatment of oscillatory disorders via the Freidlin–Wentzell quasipotential (Kriger, 2026, Section 2.2).
- (ii) The DST concept of anosognosia as distance-dependent insight decay (Eqs. 18–19 of Kriger, 2026) receives a specific mechanistic realization through the EOS sensor degradation model (Theorem 5.1), which identifies the *phase transition* in self-assessment capacity and its autocatalytic dynamics.
- (iii) The DST social isolation model (Eqs. 30–31 of Kriger, 2026) is instantiated through the EOS informational isolation condition (K5), which provides the specific network topology (star network with substance at center) and the critical one-addict principle.

- (iv) The sunk-cost entrenchment mechanism (Theorem 6.1) provides the *cognitive mechanism* underlying the progressive deepening of the potential well V_{add} in the DST framework.
- (v) The formal theory of intervention (Section 11) extends the DST treatment framework by deriving each therapeutic strategy as a structural attack on a specific EOS mechanism, with formally specified failure modes and synergy structures.

Section 10 provides a formal correspondence table mapping all EOS variables to the DST state-space representation.

1.6 Empirical Foundation

The empirical observations underpinning this work derive from the author’s direct experience in addiction rehabilitation across three decades: work as an emergency medical technician in Sverdlovsk, USSR, in the 1980s; therapeutic community work in the Lifta Valley, Jerusalem, in the 1990s (including immersion in the Narcotics Anonymous 12-step framework); and operation of a residential shelter in rural Canada over a nine-year period. These experiences, documented in Kriger (2023), provide qualitative confirmation of the model’s structural predictions and motivate several key constructs.

1.7 Plan of the Paper

Section 2 establishes the agent structure and variable mapping. Section 3 develops the reward system as the exploited substrate. Section 4 formalizes the addiction cycle as a relaxation oscillator. Section 5 models the degradation of self-assessment capacity. Section 6 derives the sunk-cost entrenchment. Section 7 analyzes social environment as informational isolation. Section 8 introduces the developmental vulnerability model. Section 9 develops the neurostimulation threat model. Section 10 presents the formal correspondence with the DST framework. Section 11 presents the formal theory of intervention. Section 12 presents falsifiable predictions. Section 13 discusses implications.

2 The Agent Structure of Addiction

2.1 Reinterpretation of the Two-Agent Framework

In the general EOS framework (Kriger, 2017), the extracting agent $\mathcal{N}$ optimizes resource extraction from agent $\mathcal{W}$. In the addictive instantiation, agent $\mathcal{N}$ is not a person but the addictive substance, behavior, or device itself, modeled as an abstract agent whose “utility function” is defined by the reinforcement mechanism inherent in the agent–brain interaction.

This reinterpretation requires no modification of the mathematical structure. The substance does not *intend* anything, just as a thermostat does not intend to oscillate. The system emerges whenever the structural conditions (K1)–(K5) are satisfied, regardless of whether any component possesses intentionality. This is the *irrelevance of intent* principle (Kriger, 2017, Section 14.5).

2.2 Variable Mapping

Table 1 establishes the complete correspondence between the abstract EOS variables and three instantiations of addiction.

Table 1: Variable mapping across three addiction substrates. The abstract structure is identical; only the physical realization changes.

EOS Variable	Chemical (heroin)	Behavioral (gambling)	Neural inter- face
$E(t)$: hedonic capacity	Physical health, relationships	Financial resources, time	Neural baseline tone
$K(t)$: neuroadaptation	Opioid tolerance	Reward desensitization	Electrode–tissue impedance
τ_{int} : internal precision	Insight into dependence	Awareness of losses	Hedonic discrimination
$I(t)$: accumulated losses	Health, career, family	Money, reputation	Surgical investment, neural damage
C : social connectivity	Non-using contacts	Non-gambling friends	Pre-implant social network
L : path length	Acquisition $\rightarrow$ injection $\rightarrow$ reward	Bet $\rightarrow$ outcome $\rightarrow$ reward	Button press $\rightarrow$ stimulation

2.3 Verification of the Five EOS Conditions

(K1) Asymmetric extraction.

The addictive agent extracts resources (hedonic capacity, health, time, money, social connections) from the individual without symmetric reciprocation. The short-term hedonic reward is not reciprocation—it is the *mechanism* of extraction. The individual pays real costs; the “agent” pays nothing.

(K2) Hysteretic oscillation.

The individual alternates between use and abstinence, with switching governed by hysteretic thresholds: relapse occurs after craving exceeds a threshold for a sustained period; cessation occurs after consequences accumulate past a different threshold. The coupled system admits a stable limit cycle (Theorem 4.1).

(K3) Sensor degradation.

The individual’s capacity to accurately assess their own state is progressively eroded. In chemical addiction: denial and rationalization. In behavioral addiction: distorted time perception. In neural interface addiction: inability to distinguish stimulated from genuine well-being. This erosion produces a phase transition (Theorem 5.1).

(K4) Sunk-cost trapping.

Accumulated losses create a positive feedback loop making exit increasingly costly. The paradoxical logic: “I have already lost so much; if I quit now, all that suffering was for nothing.” The trap locks after a critical threshold (Theorem 6.1).

(K5) Informational isolation.

The addicted individual progressively restricts social contact to those who share

the addiction, creating a star-network topology. Non-using contacts are eliminated. Breaking isolation destabilizes the addictive equilibrium (Proposition 7.1).

3 The Reward System as Exploited Substrate

3.1 The Natural Reward Circuitry

The brain’s reward system operates as a feedback control loop: action $\rightarrow$ neurotransmitter release $\rightarrow$ subjective reward $\rightarrow$ motivation to repeat. The key neurotransmitters—dopamine, serotonin, endorphins, oxytocin—are released in response to survival-promoting actions (Volkow et al., 2016). The reward magnitude is proportional to effort and survival value. This proportionality is the system’s principal safeguard against runaway positive feedback.

3.2 The Hijacking Mechanism

Addictive agents bypass the natural loop by directly activating reward circuitry without requiring the survival-promoting action. This is the *shunting mechanism* (Kriger, 2023, Chapter 1): the agent exploits existing mechanisms, short-circuiting the effort-to-reward pathway.

Formally, let $R(t)$ denote the reward signal. Under natural conditions,

$$R(t) = f(a(t), s(t)), \quad (1)$$

where a is effort and s is survival relevance. Under addictive influence,

$$R(t) = g(d(t)) \gg f(a, s), \quad (2)$$

where d is dose/intensity and g is the agent’s activation function. The critical feature: g is independent of a and s . Reward is decoupled from effort and survival value.

3.3 The Inverse Proportionality Principle

Principle 3.1 (Inverse Proportionality). *The stability of a reward-driven system is inversely proportional to the directness of the pathway from action to reward.*

Natural rewards require effort, time, and engagement with the external world. Addictive agents provide the same neurochemical result with minimal effort. The shorter the path, the more catastrophic the instability. This explains the addictive potential ranking: heroin $>$ cocaine $>$ alcohol $>$ cannabis. It also predicts that direct neural stimulation, which eliminates the pathway entirely, will be maximally addictive.

3.4 The Olds–Milner Theorem

Theorem 3.2 (Self-Destruction under Direct Reward Access). *If an agent is given direct access to maximization of its reward signal, bypassing all intermediary functions, the agent will maximize reward to the exclusion of all other needs, resulting in self-destruction.*

The Olds–Milner experiment (Olds & Milner, 1954)—rats pressing a lever for direct brain stimulation until death—is the empirical existence proof. The formal framework provides the structural explanation. This theorem is substrate-independent and applies to any system with a short-circuited reward pathway.

4 The Addiction Cycle as Relaxation Oscillator

4.1 State Variables and Governing Equations

Let $E(t)$ denote hedonic/functional capacity (depleted by use, replenished during abstinence) and $K(t)$ denote neuroadaptation/tolerance (increases during use, decays slowly during abstinence). The governing equations are:

During use:

$$\frac{dE}{dt} = \alpha_0 - \beta_0 E - \gamma(K) \cdot E, \quad (3)$$

During abstinence:

$$\frac{dE}{dt} = \alpha_1 - \beta_1 E, \quad (4)$$

Tolerance dynamics:

$$\frac{dK}{dt} = \delta E - \varepsilon K, \quad (5)$$

where α_0 is hedonic input rate during use, β_0 is natural decay, $\gamma(K)$ is tolerance-dependent depletion (increasing in K), $\alpha_1 < \alpha_0$ is natural recovery rate, β_1 is withdrawal-period decay, δ governs tolerance accumulation, and ε governs tolerance decay. For neural interface addiction, $\gamma(K)$ also includes electrode–tissue interface degradation.

4.2 Hysteretic Switching and the Limit Cycle

Switching between modes occurs at thresholds E_L (relapse: craving overcomes resistance) and E_H (cessation: consequences exceed tolerance). The hysteretic structure ($E_L < E_H$; system remains in current mode for $E_L < E < E_H$) follows from optimal switching theory (Filippov, 1988).

Theorem 4.1 (Existence of the Addiction Limit Cycle). *Under conditions*

(H1) $\alpha_1 > \beta_1 E_L$;

(H2) $E_L < E_H$, both interior;

(H3) nullcline intersection conditions;

the system (3)–(5) admits a stable limit cycle in the (E, K) phase plane, corresponding to the binge–remission–tension–relapse cycle.

Proof sketch. Construct trapping region, verify inward-pointing vector field, confirm absence of fixed points in the interior, apply the Poincaré–Bendixson theorem via Filippov theory for piecewise-smooth systems (Filippov, 1988; di Bernardo et al., 2008). The full proof is identical to (Kriger, 2017, Theorem 4.1). $\square$

Remark 4.2 (Period). *The oscillation period is approximately*

$$T \approx \frac{1}{\beta_1} \ln \frac{E_H}{E_L} + \frac{1}{\beta_0 + \gamma} \ln \frac{E_H}{E_L}. \quad (6)$$

Higher depletion rates produce shorter cycles. For neural interface addiction, extremely high $\gamma(K)$ predicts cycles on the order of minutes to hours.

Remark 4.3 (Relationship to DST limit-cycle models). *In the DST framework (Kriger, 2026), oscillatory disorders (bipolar disorder, OCD) are modeled as limit cycles with the potential landscape replaced by the Freidlin–Wentzell quasipotential $\tilde{V}$. The addiction limit cycle (3)–(5) is a piecewise-smooth relaxation oscillator, for which the quasipotential is defined piecewise on each mode, with transition rates governed by the Kramers escape formula applied to the quasipotential barriers between modes. This provides the formal bridge: the “depth” of the addiction attractor in DST corresponds to the amplitude and robustness of the limit cycle in EOS.*

5 Degradation of Self-Assessment Capacity

5.1 The Epistemic Model

The addicted individual maintains a belief μ about their true state (“How badly am I affected?”). They receive internal signals (own experience) with precision τ_{int} and external signals (addictive environment: peers, rationalization, substance’s euphoric effect) with precision τ_{ext} . The key mechanism:

$$\Delta\tau_{\text{int}} = -\eta \cdot \mathbf{1}\{\text{accept external signal over internal}\}. \quad (7)$$

Each acceptance of the contradictory external narrative permanently reduces internal sensing confidence.

Theorem 5.1 (Phase Transition in Self-Assessment). *There exists a critical time*

$$t^* = \frac{\tau_{\text{int}}(0)}{2\eta} \quad (8)$$

such that for $t > t^$, self-assessment is predominantly determined by external signals. The transition at t^* is sharp (phase transition), not gradual, due to the autocatalytic positive feedback: lower $\tau_{\text{int}} \rightarrow$ higher acceptance probability $\rightarrow$ further reduction of τ_{int} .*

For neural interface addiction, this takes the form of hedonic confusion: the individual cannot distinguish stimulation-induced well-being from genuine well-being. The phase transition may occur within weeks rather than years.

5.2 Relationship to the DST Anosognosia Model

The DST framework (Kriger, 2026) models anosognosia through a distance-dependent insight function:

$$\varphi(d) = \varphi_0 d \exp\left(-\frac{d^2}{2\delta^2}\right), \quad (9)$$

where $d = \|x - x_H^*\|$ is the distance from the healthy attractor. For small d , insight is proportional to deviation; for large d , insight vanishes (“the attractor does not know it is an attractor”).

The EOS sensor degradation model provides the *mechanism* underlying this distance dependence. Along the addiction trajectory, the distance $d(t)$ from the healthy state increases monotonically (as E decreases and K increases). The EOS phase transition at t^* corresponds to the moment when $d(t^*)$ exceeds the DST insight horizon δ :

$$d(t^*) \approx \delta \iff \tau_{\text{int}}(t^*) = \frac{\tau_{\text{int}}(0)}{2}. \quad (10)$$

Before this threshold, the individual has partial insight (DST: $\varphi(d) > 0$; EOS: $\tau_{\text{int}} > \tau_{\text{ext}}$). After it, insight is effectively lost (DST: $\varphi(d) \approx 0$; EOS: $\tau_{\text{int}} \ll \tau_{\text{ext}}$).

5.3 Empirical Confirmation

The rehabilitation center phrase “We love you, we hate your disease” reflects the separation between the person (whose sensor has been degraded) and the system (which performed the degradation). The experience of encountering addicts who appeared as “empty shells” (Kriger, 2023, Chapter 3) corresponds to the post-transition regime $\tau_{\text{int}} \approx 0$: autonomous self-assessment is no longer possible.

6 The Sunk-Cost Entrenchment Mechanism

Let $I(t)$ represent accumulated losses and $D(t)$ represent cognitive dissonance. Resolution proceeds via two channels:

Exit: Cost $C_{\text{exit}} = \phi(I) + \psi(\tau_{\text{int}})$, increasing in I , decreasing in τ_{int} .

Reinterpretation: Cost $C_{\text{stay}} = \xi/r(t)$, decreasing as rationalization capacity r grows.

Theorem 6.1 (Investment Trap). *Under $dI/dt = r(t)$ and $dr/dt = \zeta I$:*

- (i) $I(t)$ is strictly increasing and unbounded;
- (ii) there exists I_{cr} such that for $I > I_{\text{cr}}$, $C_{\text{exit}} > C_{\text{stay}}$ permanently;
- (iii) the system is self-reinforcing past I_{cr} .

The paradoxical form in addiction: “I have already lost my health, my family, my career. If I quit now, all that suffering was for nothing.” Breaking the trap requires decoupling exit cost from accumulated investment—a specific structural intervention (Section 11.2).

Remark 6.2 (Relationship to DST attractor deepening). *In the DST framework (Kriger, 2026, Appendix G), the addiction potential deepens according to*

$$\frac{d\Lambda_{\text{add}}}{dt} = \kappa_{\text{add}} x_{\text{use}}(t) - \gamma_{\text{add}} \Lambda_{\text{add}}. \quad (11)$$

The EOS investment trap provides the cognitive mechanism for this deepening: it is not only the neuroadaptation (Λ_{add}) that deepens the well, but also the sunk-cost logic ($I \rightarrow C_{\text{exit}} \uparrow$) that makes escape subjectively costlier. The DST quantity Λ_{add} should be understood as the sum of neuroadaptive and cognitive entrenchment:

$$\Lambda_{\text{total}}(t) = \Lambda_{\text{neuro}}(t) + \Lambda_{\text{cog}}(I(t)), \quad (12)$$

where Λ_{cog} is an increasing function of accumulated losses.

7 Social Environment as Informational Isolation

7.1 The Star Network and Its Disruption

The addicted individual’s social network reorganizes into a star topology with the substance at the center. Non-using contacts are eliminated.

Proposition 7.1 (Equilibrium Destabilization). *If the individual receives corrective information from outside the addictive network of sufficient magnitude to reduce effective exit cost below accommodation cost, the addictive equilibrium is destroyed and exit becomes preferred.*

7.2 The One-Addict Principle

Empirical observation (Kriger, 2023, Chapter 3): a single addict placed among non-users could remain clean for months. Introduction of a second addict almost invariably triggered relapse in both. This is a direct consequence of (K5): the second addict reopens the information channel to the addictive agent. Cross-substance contamination (romantic relationships with alcoholics triggering relapse in heroin addicts) is explained by the same mechanism: the partner reintroduces the star topology with a substance at the center.

7.3 The Contagion Dynamics

The pro-use signal from an active or recovering addict has anomalously high persuasive precision. In Bayesian terms, the signal-to-noise ratio of the pro-use message is engineered (by years of addictive practice) to overwhelm even strong priors against use. The author reports that a recovering addict’s enthusiastic suggestion to use drugs was momentarily persuasive even to a clergyman who had never used substances (Kriger, 2023).

7.4 Correspondence with DST Social Isolation Dynamics

The DST framework models social connectivity as a stabilizing feedback (Kriger, 2026, Eqs. 30–31):

$$\frac{dC}{dt} = -\beta_C(x_H^* - x)^2 + \gamma_C(C_0 - C), \quad (13)$$

where worsening mental state drives social withdrawal. The EOS star-network model provides the *topology* of this withdrawal: it is not a uniform reduction in connectivity but a selective elimination of corrective contacts while maintaining (or strengthening) connections that support the addictive equilibrium. The one-addict principle identifies a *discontinuity* in the DST connectivity variable C : the introduction of a single co-addict qualitatively restructures the information environment.

8 The Developmental Vulnerability Window

8.1 The Endorphin Decline Model

Children experience elevated baseline reward hormones relative to body mass, providing a biochemical buffer against childhood stresses (Kriger, 2023, Chapter 2). This baseline declines during adolescence:

$$E_0(\text{age}) = E_{\text{child}} \cdot \exp(-\lambda(\text{age} - \text{age}_{\text{onset}})) \quad \text{for } \text{age} > \text{age}_{\text{onset}}, \quad (14)$$

where λ is the decline rate and $\text{age}_{\text{onset}} \approx 12\text{--}13$. The decline reflects an evolutionary mechanism: reduced endorphin support motivates the search for external satisfaction (social bonds, partners, achievement) serving reproductive goals (Piloizzi et al., 2021).

8.2 The Vulnerability Window

Self-regulation capacity develops on a separate, delayed trajectory:

$$S(\text{age}) = S_{\text{adult}} \cdot (1 - \exp(-\mu(\text{age} - \text{age}_{\text{onset}}))). \quad (15)$$

The critical observation: $\mu < \lambda$. Self-regulation matures more slowly than the endorphin baseline declines. The gap

$$V(\text{age}) = \frac{E_0(\text{onset})}{E_0(\text{age})} - \frac{S(\text{age})}{S_{\text{adult}}} \quad (16)$$

is positive during approximately ages 13–25, peaking around 15–18. During this window, the individual has declining natural reward capacity combined with immature self-regulation—maximal susceptibility to system capture.

Theorem 8.1 (Adolescent Vulnerability). *Capture probability is monotonically increasing in $V(\text{age})$. Positive alternatives narrow the window by increasing S and/or maintaining E_0 .*

9 Neural Interface Addiction: The Limiting Case

9.1 The Technology

Neural interface technologies aim to establish direct communication between electronic devices and brain tissue (Musk & Neuralink, 2019). Current medical applications (motor restoration, depression treatment, pain management) use the same pathways that could stimulate the nucleus accumbens for pleasure. We are not discussing science fiction: the Olds–Milner experiment demonstrated direct reward stimulation in 1954; deep brain stimulation targets reward-adjacent circuits today.

9.2 The Zero-Path-Length Limit

Table 2 summarizes the path length from action to reward across addiction substrates.

Table 2: Path length from action to reward. The inverse proportionality principle (Principle 3.1) predicts instability increases as L decreases.

Substrate	Path length L	Typical cycle period
Natural reward	Long (effort $\rightarrow$ achievement $\rightarrow$ reward)	N/A (no pathological cycle)
Alcohol	Medium (purchase $\rightarrow$ consumption $\rightarrow$ absorption)	Weeks to months
Heroin (IV)	Short (injection $\rightarrow$ seconds to reward)	Days to weeks
Gambling (digital)	Very short (click $\rightarrow$ outcome)	Hours to days
Neural interface	Zero (button $\rightarrow$ immediate stimulation)	Minutes to hours (predicted)

9.3 Predictions for Neural Interface Addiction

(N1) Ultra-Short Cycles.

Period T on the order of minutes to hours, not weeks to months, due to rapid neural adaptation.

(N2) Accelerated Sensor Degradation.

Phase transition time t^* dramatically shorter; hedonic confusion may onset within weeks.

(N3) Immediate Sunk-Cost Lock.

Surgical investment creates $I(0)$ that may exceed I_{cr} at implantation. The trap locks before the first stimulation session.

(N4) Novel Failure Modes.

Electrode degradation, infection, device malfunction, and qualitatively new withdrawal upon device removal.

9.4 The Black Market Scenario

The framework predicts with high confidence the emergence of “jailbroken” neural interfaces reprogrammed for direct pleasure stimulation. This market would be more dangerous than the existing drug trade in several respects: no chemical trace (detection-resistant), higher addictive potential than any known substance, extreme surgical risk, and immediate sunk-cost lock. Current drug policy frameworks are wholly inadequate.

However, a balanced analysis must acknowledge that neural interface addiction also *eliminates* several categories of harm inherent in chemical substance use:

- (i) **No injection-related harm.** The entire epidemiology of needle-borne disease—HIV, hepatitis B and C, bacterial endocarditis, soft-tissue infections—is eliminated. For intravenous drug users, these secondary harms often exceed the direct pharmacological damage.
- (ii) **No toxic metabolites or organ damage.** Chemical substances impose dose-dependent toxicity on the liver, kidneys, cardiovascular system, and lungs. Neural stimulation, by contrast, produces no circulating toxins and no metabolic burden.
- (iii) **No withdrawal syndrome in the classical sense.** The agonizing physiological withdrawal characteristic of opioids, alcohol, and benzodiazepines—which is itself a major source of medical emergencies and a powerful driver of continued use—is absent. The “withdrawal” from neural stimulation is psychological (anhedonia, craving) rather than physiological (seizures, autonomic crisis).
- (iv) **No recurring procurement cost.** Unlike chemical substances, which require continuous purchase and expose the user to criminal markets, violence, and adulterated product, a neural interface is a one-time acquisition. The user is freed from the procurement cycle and its associated harms: no dealer interactions, no financial ruin from daily purchases, no risk of overdose from variable purity.
- (v) **Dose precision and controllability.** Stimulation parameters (intensity, frequency, duration) can in principle be precisely controlled and logged, unlike chemical doses which vary with purity, route, and individual metabolism. This creates the theoretical possibility of harm-reduction protocols impossible with street drugs.

These advantages do not diminish the fundamental danger identified by the framework—the $L = 0$ path length makes neural stimulation maximally addictive (Principle 3.1)—but

they alter the *harm profile* qualitatively. The policy implication is nuanced: neural interface addiction may produce fewer secondary medical casualties per user than heroin or methamphetamine, while producing *more* addicts and deeper behavioral capture. The net public health impact depends on the relative weights of these opposing effects—a question the present framework identifies but cannot resolve without empirical data.

10 Formal Correspondence with the DST Framework

Table 3 provides the complete mapping between EOS variables and the DST state-space representation of Kriger (2026). This correspondence establishes that the two frameworks are mutually consistent: the DST framework provides the global geometry (attractor landscape, basin boundaries, resilience measures), while the EOS framework provides the internal mechanics (oscillatory dynamics, sensor degradation kinetics, cognitive entrenchment, network topology).

Table 3: Formal correspondence between EOS and DST frameworks.

EOS construct	DST construct	Relationship
Limit cycle in (E, K)	Disorder attractor x_D^*	Limit cycle is internal dynamics within the DST attractor basin; quasipotential $\tilde{V}$ replaces V
Sensor degradation $(\tau_{\text{int}} \rightarrow 0)$	Anosognosia $(\varphi(d) \rightarrow 0)$	Phase transition at t^* corresponds to $d(t^*) = \delta$
Sunk-cost trap $(I > I_{\text{cr}})$	Attractor deepening $(\Lambda_{\text{add}} \uparrow)$	Cognitive entrenchment adds to neuroadaptive deepening: $\Lambda_{\text{total}} = \Lambda_{\text{neuro}} + \Lambda_{\text{cog}}(I)$
Star-network isolation (K5)	Social connectivity $C(t) \downarrow$	Star topology specifies the <i>structure</i> of connectivity loss
Hysteretic thresholds E_L, E_H	Hysteresis: $s_{\text{onset}} > s_{\text{offset}}$	Thresholds map to barrier heights at basin boundaries
Tolerance $K(t)$	Allostatic shift $x_{H,\text{eff}}^*$	$x_{H,\text{eff}}^* = x_H^* - \tau_{\text{tol}} x_{\text{tol}}$ (Kriger, 2026, Eq. G.4)
Path length L	(not modeled in DST)	Novel EOS contribution: L determines instability magnitude
Vulnerability gap $V(\text{age})$	Resilience parameter μ	$V \text{ high} \Leftrightarrow \mu \text{ low} \Leftrightarrow$ shallow healthy basin
Intervention loops 1–3	Therapeutic operations on V, W, C	Each loop targets specific DST parameters

The correspondence reveals two classes of predictions:

- (a) **Predictions generated by both frameworks independently.** Hysteresis (recovery harder than onset), self-sustaining dynamics after stressor removal, social isolation as positive feedback. These provide mutual validation.
- (b) **Predictions generated by one framework but not the other.** The EOS framework uniquely predicts: the phase-transition structure of sensor degradation (sharp, not gradual), the sunk-cost lock threshold I_{cr} , the one-addict principle (discontinuous relapse), and the path-length dependence of instability. The DST framework

uniquely predicts: critical slowing down signatures before the initial transition, the $3/2$ -power scaling law for resilience near the tipping point (Kriger, 2026, Eq. 14), and the fear-of-madness autocatalytic loop (Kriger, 2026, Eqs. 20–22). The latter is relevant to addiction as the “fear of relapse” loop—a metacognitive anxiety that can itself precipitate relapse by eroding the resilience of the recovery state.

11 A Formal Theory of Intervention

11.1 From Clinical Intuition to Structural Engineering

The EOS framework transforms intervention design from intuition-guided practice to structural engineering. Each theorem identifies a specific mechanism sustaining the addictive equilibrium. Each mechanism has a mathematically defined vulnerability. Each vulnerability admits a class of interventions whose mechanism of action, predicted efficacy, failure modes, and synergies with other interventions can be formally specified.

We organize the theory around the three coupled feedback loops identified in Kriger (2017, Section 14.3), extended with the developmental and neurostimulation dimensions. Each loop is a distinct target, and breaking any single loop weakens the entire system. Breaking two loops simultaneously produces effects that are more than additive. Breaking all three effectively destroys the EOS structure.

11.2 Loop 1: The Investment–Rationalization Loop

Mechanism: More invested $\rightarrow$ costlier to acknowledge loss $\rightarrow$ rationalization $\rightarrow$ continued investment $\rightarrow$ further accumulation. This is the engine of Theorem 6.1.

Structural vulnerability: The loop depends on the coupling between accumulated investment I and perceived exit cost C_{exit} . If this coupling is broken—if the individual can be brought to perceive exit cost as independent of what has already been lost—the loop collapses.

Intervention class 1A: Investment reframing.

Cognitive interventions that explicitly decouple exit cost from sunk cost (Festinger, 1957). The formal model predicts that this intervention is most effective before I crosses I_{cr} , and nearly ineffective after.

Intervention class 1B: Exit cost reduction.

Practical support that reduces C_{exit} directly, regardless of I . Housing assistance, vocational training, legal aid, financial support, childcare. Each unit of direct exit cost reduction is equivalent to raising the threshold I_{cr} —buying time before the trap locks.

Intervention class 1C: Investment amnesty.

For neural interface addiction, where $I(0)$ may exceed I_{cr} from surgical investment alone: public health programs guaranteeing free, safe device removal. This is the structural equivalent of resetting I to zero.

Failure mode prediction: Interventions targeting Loop 1 alone will fail if Loops 2 and 3 remain intact.

11.3 Loop 2: The Precision Erosion–Acceptance Loop

Mechanism: Each acceptance of contradictory external narrative $\rightarrow$ reduced internal precision $\tau_{\text{int}} \rightarrow$ higher probability of future acceptance $\rightarrow$ further erosion. This is the autocatalytic process of Theorem 5.1.

Structural vulnerability: The loop depends on the monotonic decline of τ_{int} . If τ_{int} can be stabilized or increased, the autocatalytic process is interrupted.

Intervention class 2A: Precision restoration (pre-transition).

Before t^* , the individual’s internal sensor is damaged but functional. Interventions that rebuild τ_{int} include: psychotherapy focused on reality testing; journaling and self-monitoring; trusted external feedback from non-addicted individuals. The model predicts that effectiveness declines sharply as $t \rightarrow t^*$, and is near zero after.

Intervention class 2B: External sensor substitution (post-transition).

After t^* , the individual’s internal sensor is effectively destroyed. The only effective intervention is substitution of an external sensor: a trusted person, a therapist, a structured assessment tool. A sponsor in a 12-step program functions as an external sensor (Kriger, 2023).

Intervention class 2C: Hedonic confusion monitoring (neural interfaces).

For neural interface addiction: periodic external assessment using objective neurological measures (not self-report).

Failure mode prediction: Precision restoration fails if attempted post-transition or if the external sensor is compromised.

11.4 Loop 3: The Isolation–Dependence Loop

Mechanism: Absence of corrective information $\rightarrow$ dependence on addictive environment as sole information source $\rightarrow$ further isolation from corrective contacts. This is condition (K5).

Intervention class 3A: Environmental relocation.

Physical removal from the addictive environment. The author’s experience (Kriger, 2023) confirms that geographic isolation from the addictive network was sufficient to maintain abstinence for months.

Intervention class 3B: The one-addict principle.

The optimal configuration: place one recovering addict in a community of non-users. Never two. Two addicts create a mutual information channel that re-establishes the star topology. The model predicts that relapse probability increases *discontinuously* when a second addict is introduced.

Intervention class 3C: Network illumination.

Breaking informational isolation by connecting the individual to others who have *exited* similar addictions. Support groups, mentorship by former addicts, peer networks.

Intervention class 3D: Counter-signal infrastructure.

Institutional design creating persistent corrective signals: public health messaging, school-based programs, workplace wellness programs. The model predicts that effectiveness depends on signal precision relative to the addictive agent’s precision.

11.5 Combined Interventions: The Three-Loop Attack

The formal model’s most powerful prediction is that simultaneous attack on multiple loops produces synergistic effects.

Two-loop combination: Breaking Loop 3 (relocation) + Loop 2 (external sensor) is the structural description of the therapeutic community model (Kriger, 2023). The model predicts this is effective precisely because it attacks two loops simultaneously.

Three-loop combination: Breaking all three loops requires: (a) environmental relocation or network illumination (Loop 3); (b) reality testing via trusted external sensor (Loop 2); (c) investment reframing plus practical exit support (Loop 1). This is the formal justification for comprehensive rehabilitation programs.

11.6 The Vulnerability Window Interventions

Theorem 8.1 identifies a distinct intervention target: narrowing the developmental gap $V(\text{age})$ before system capture occurs.

Intervention class 4A: Endogenous reward maintenance.

Physical activity, creative expression, social engagement sustain the natural reward baseline E_0 (Piloizzi et al., 2021).

Intervention class 4B: Self-regulation acceleration.

Mentorship, mindfulness training, structured decision-making experiences accelerate $S(\text{age})$.

Intervention class 4C: Environmental prophylaxis.

Maintaining the adolescent’s environment free of addictive triggers. The critical caveat (Kriger, 2023): excessive restriction can increase attraction. The optimal strategy is restriction combined with positive alternatives.

11.7 The “Just for Today” Principle: A Formal Interpretation

The Narcotics Anonymous principle of “just for today” has a precise formal interpretation: it prevents the investment variable I from crossing I_{cr} in the reverse direction. By reframing sobriety as a daily commitment rather than a permanent one, the individual avoids the psychological weight of lifelong commitment. The principle resets the phase counter at each dawn, keeping the effective I within a single-day bound and preventing the sunk-cost trap from operating on the recovery side.

11.8 Methadone and Substitution Therapy: A Formal Interpretation

Methadone therapy replaces the addictive agent with a controlled substitute that satisfies condition (K1) more weakly: the extraction rate is lower, the oscillation amplitude is smaller, and the sensor degradation is slower. Formally, methadone maintains the individual within the limit cycle but reduces all parameters (α_0 , γ , η) to less destructive values. The cycle continues, but at lower amplitude and longer period.

The model explains both why methadone works (it reduces harm by moderating the oscillator) and why critics object (it does not break the oscillator; it merely domesticates

it). The model also predicts the conditions under which methadone *combined* with loop-breaking interventions can terminate the cycle entirely.

11.9 Intervention Design for Neural Interface Addiction

Regulatory prevention (pre-emptive).

Because (N3) predicts immediate sunk-cost lock, the primary intervention must prevent system capture: strict regulation, mandatory long-term monitoring, and legal/technical frameworks guaranteeing device removability.

Hardware safeguards.

Hard-coded limits on stimulation intensity, frequency, and duration—enforced at the hardware level, not overridable by software—artificially preserve the effort-to-reward proportionality.

Hedonic confusion monitoring.

Periodic external assessment more frequently than $1/t^*$.

Public health infrastructure for device removal.

Intervention class 1C: publicly funded programs for safe device removal, established *before* the technology becomes widespread.

11.10 Summary: The Intervention Taxonomy

Table 4 summarizes the complete intervention taxonomy.

Table 4: Complete intervention taxonomy derived from the formal model. Each intervention targets a specific theorem, loop, or parameter.

Class	Target	Mechanism	Predicted failure mode
1A	Loop 1: $I \leftrightarrow C_{\text{exit}}$	Decouple sunk cost from exit cost	Ineffective if $I > I_{\text{cr}}$
1B	Loop 1: C_{exit} directly	Reduce objective exit cost	Insufficient alone if Loops 2,3 intact
1C	Loop 1: $I(0)$ (neural)	Reset investment to zero	Requires surgical infrastructure
2A	Loop 2: τ_{int} (pre- t^*)	Rebuild internal precision	Ineffective post-transition
2B	Loop 2: substitute sensor	External calibration	Fails if sensor compromised
2C	Loop 2: neural monitoring	Objective neurological measures	Requires clinical infrastructure
3A	Loop 3: (K5)	Environmental relocation	Fails if contaminated
3B	Loop 3: topology	One-addict principle	Fails with second addict
3C	Loop 3: information	Network illumination	Fails if group infiltrated
3D	Loop 3: signal	Counter-signal infrastructure	Fails if perceived as in-authentic
4A	$V(\text{age}): E_0$	Maintain endogenous reward	Requires sustained engagement
4B	$V(\text{age}): S$	Accelerate self-regulation	Most effective during window
4C	$V(\text{age}):$ environment	Environmental prophylaxis	Excessive restriction backfires

12 Falsifiable Predictions

Prediction 1 (Critical Investment Threshold). *There exists I_{cr} such that exit probability drops discontinuously as accumulated loss crosses I_{cr} . Test: longitudinal study of addiction duration, losses, and exit probability. Sharp transition predicted, not gradual decline.*

Prediction 2 (Oscillation Period). *Binge-relapse cycle period T is computable from substance depletion rate, recovery rate, and craving threshold via Eq. (6). Test: fit period formula to relapse intervals across substances.*

Prediction 3 (Self-Assessment Phase Transition). *Self-reported judgment confidence undergoes a phase transition as a function of addiction duration. Test: bimodal distributions in cross-sectional data; acceleration in longitudinal data.*

Prediction 4 (Second-Addict Effect). *Introduction of a second addict into a recovery environment increases relapse probability discontinuously. Test: controlled comparison of single vs. dual settings.*

Prediction 5 (Neural Interface Risk). *If/when neural stimulation devices become available: (a) higher addictive potential than any chemical substance; (b) shorter cycle periods; (c) faster sensor degradation; (d) immediate sunk-cost trapping.*

Prediction 6 (Intervention Synergy). *Simultaneous two-loop interventions produce effects exceeding the sum of single-loop interventions. Test: compare outcomes of comprehensive programs targeting identified loops vs. single-component programs.*

Prediction 7 (Pre- vs. Post-Transition Intervention). *Precision restoration (class 2A) is effective only before the phase transition at t^* . After t^* , only external sensor substitution (class 2B) produces measurable improvement. Test: stratify intervention outcomes by estimated time since addiction onset relative to predicted t^* .*

Prediction 8 (DST–EOS Joint Predictions). *The following predictions arise from the conjunction of both frameworks:*

- (a) *Critical slowing down signatures (rising autocorrelation and variance in mood/craving time series) should precede the initial transition into the addiction limit cycle, with the scaling predicted by the DST Ornstein–Uhlenbeck model (Kriger, 2026, Eq. 11).*
- (b) *The “fear of relapse” in recovery operates as an autocatalytic destabilizing loop (analogous to the fear-of-madness model of Kriger, 2026, Eqs. 20–22), reducing the resilience of the recovery attractor and accelerating return to the addiction limit cycle.*
- (c) *Recovery hysteresis is quantifiable: the stress reduction required to exit the addiction attractor exceeds the stress level that initiated the transition, with the asymmetry predicted by barrier height calculations from both frameworks.*

Falsification. The framework is falsified if addiction is non-oscillatory, produces no sensor degradation, exhibits no investment trap, is independent of social environment, or if multi-loop interventions are no more effective than single-loop interventions.

13 Discussion and Conclusion

13.1 Summary

This paper has developed a rigorous, substrate-independent formal theory of addiction as an instantiation of the extractive oscillator with sensor degradation. The principal contributions are: (1) full formal apparatus with variable mappings, equations, and theorems; (2) the developmental vulnerability model; (3) the neurostimulation threat analysis; (4) a formal theory of intervention in which every strategy has a mathematically specified target, mechanism, failure mode, and synergy structure; (5) formal correspondence with the DST framework for psychiatric disorders (Kriger, 2026); (6) eight falsifiable predictions, including joint DST–EOS predictions.

13.2 Empirical Support from the Diversity of Addictions

The strongest empirical argument for the substrate-independent EOS framework is the *existence of behavioral addictions*. If addiction were fundamentally a neurochemical phenomenon—a property of specific molecules interacting with specific receptors—then

gambling disorder, gaming disorder (ICD-11), social media compulsion, and pornography addiction would be inexplicable. No exogenous molecule enters the body. No receptor is directly activated by an external ligand. Yet the phenomenology is identical: the oscillatory binge–remission–relapse cycle (K2), the progressive loss of self-assessment (“I can stop whenever I want”) (K3), the sunk-cost entrenchment (“I’ve already lost so much, one more bet might win it back”) (K4), and the social isolation into communities of fellow users (K5).

This convergence across radically different substrates is precisely what the EOS framework predicts and what substrate-specific theories cannot explain without ad hoc extensions. Consider the range:

- (i) **Gambling disorder** (DSM-5-TR): no exogenous substance; reward delivered via intermittent reinforcement schedules (Ferster & Skinner, 1957). Path length L is short (bet $\rightarrow$ outcome) but nonzero. All five EOS conditions are satisfied. The sunk-cost trap is particularly vivid: the gambler’s fallacy (“I’m due for a win”) is a specific instantiation of the rationalization function $r(t)$ in Theorem 6.1.
- (ii) **Gaming disorder** (ICD-11): reward via achievement signals, social status within game, and variable-ratio reinforcement. Sensor degradation manifests as inability to perceive time spent or real-world consequences. Isolation (K5) takes the form of replacement of offline social networks with in-game communities.
- (iii) **Social media compulsion**: reward via dopaminergic response to social validation (likes, comments, followers). The platform’s algorithmic design *optimizes* extraction (K1) using data-driven personalization—the first case where the “extracting agent” possesses genuine computational intelligence, unlike the passive chemical agent. Path length is extremely short: scroll $\rightarrow$ dopamine.
- (iv) **Pornography addiction**: supernormal stimulus (Robinson & Berridge, 2003) producing tolerance (escalation to more extreme content = $K(t)$ increasing), sensor degradation (distorted perception of real relationships), and social isolation.
- (v) **Work addiction (workaholism)**: reward via achievement, status, and avoidance of emotional distress. Uniquely, this addiction is often *socially rewarded*, making the isolation condition (K5) harder to satisfy—which the framework correctly predicts should make it less entrapping than socially stigmatized addictions.
- (vi) **Food addiction**: despite involving ingestion, the addictive mechanism for ultra-processed food is not pharmacological in the traditional sense but operates through supernormal reward signals (sugar–fat–salt combinations that exceed any natural food stimulus).

The fact that identical dynamical structures—oscillation, sensor degradation, sunk-cost trapping, isolation—emerge across substrates with *no shared chemistry* constitutes strong evidence that the governing mechanism is structural, not chemical. The EOS conditions (K1)–(K5) identify the abstract invariant: any system satisfying these five conditions will exhibit addictive dynamics, regardless of whether the reward pathway involves μ -opioid receptors, dopamine D2 receptors, or no pharmacology at all.

This observation also has direct implications for the DST framework (Kriger, 2026): the addiction attractor in the potential landscape is not a single well but a *family of wells*

with identical topology but different physical substrates. Cross-addiction (substitution of one addictive behavior for another upon cessation) is explained naturally: the individual exits one well but, if the structural conditions (K1)–(K5) remain satisfied by an alternative substrate, enters an isomorphic well nearby in state space. Durable recovery requires addressing the structural conditions themselves, not merely removing one specific substrate.

13.3 The Deepest Structural Insight

Addiction is not a disease of chemistry, personality, or willpower. It is a structural property of any system in which an agent can short-circuit the natural effort-to-reward pathway. The five conditions (K1)–(K5) are not specific to any substance or behavior. They describe the abstract geometry of extraction. Eliminating heroin does not eliminate the conditions; it redirects the system to another substrate. The only durable approach addresses the structural conditions: maintain endogenous reward (keep E_0 high), accelerate self-regulation (increase S), break isolation (violate K5), reduce exit costs (decouple I from C_{exit}), and preserve effort-to-reward proportionality (keep L large).

13.4 Integration with the DST Framework

The correspondence established in Section 10 demonstrates that the EOS and DST frameworks are not competing theories but complementary levels of description. The DST framework provides the *macroscopic geometry*: the potential landscape on which the healthy and addicted states coexist as attractors, the resilience measure that quantifies vulnerability, the critical slowing down that precedes transitions, and the hysteresis that makes recovery harder than prevention. The EOS framework provides the *microscopic mechanics*: the relaxation oscillator that generates the binge–remission cycle, the epistemic model that explains the loss of self-assessment, the cognitive mechanism that deepens the trap, and the network topology that maintains isolation.

Together, they generate predictions that neither framework produces alone (Prediction 8). The “fear of relapse” as an autocatalytic destabilizing loop—directly analogous to the DST fear-of-madness model—is a clinically significant prediction that emerges only from the integration: the DST framework provides the mathematical structure (effective resilience reduction via anxiety coupling), while the EOS framework identifies the specific context (recovery from addiction) in which this loop is most dangerous.

13.5 The Urgency of the Neural Interface Question

The analysis of Section 9 constitutes the first formal risk assessment of neural interface addiction grounded in a mathematical framework. The technology combines maximal addictive potential ($L = 0$), immediate sunk-cost trapping ($I(0) > I_{\text{cr}}$), and fastest sensor degradation (hedonic confusion). Current regulatory frameworks are wholly inadequate. The Olds–Milner experiment was conducted in 1954 (Olds & Milner, 1954). Seventy years later, the means to reproduce it in humans are being actively developed.

13.6 The Intervention as Engineering

The formal theory of intervention (Section 11) represents a paradigm shift from clinical art to structural engineering. Every known intervention maps to a formal target. Unknown

interventions can be predicted by identifying unattacked vulnerabilities. The effectiveness of combined interventions can be predicted from the synergy structure. The timing constraints (before I_{cr} , before t^*) are derivable from the model’s phase-transition structure. This does not replace clinical judgment; it provides the formal scaffolding within which clinical judgment operates with greater precision and accountability.

13.7 Ethical Note

Reducing human suffering to equations risks dehumanization. We contend that formal understanding enables more effective compassion: knowing the mechanism of a trap is the first step toward designing an escape. The individuals whose stories appear in Kriger (2023)—the heroin addicts in the Lifta Valley, the recovering users in the Canadian shelter, the women trapped in cycles of violence and alcoholism—are not data points. They are people. The equations describe their prison. The interventions describe the keys.

13.8 Limitations and Future Directions

Model parameters require empirical calibration. Qualitative results are robust to functional form, but quantitative predictions depend on parameters obtainable only through longitudinal studies. The neural interface predictions are prospective. The developmental vulnerability model requires cross-cultural and cross-species testing. Extension to multi-substance addiction, process addictions, and chemical-behavioral interaction remain future directions. The formal theory of intervention requires empirical validation through controlled trials designed to test specific loop-targeting predictions.

The integration with the DST framework opens additional research directions: empirical measurement of critical slowing down before addiction onset (using ecological momentary assessment), formal modeling of the fear-of-relapse loop using the DST metacognitive anxiety equations, and cross-disorder analysis of how the addiction attractor interacts with comorbid disorder attractors (depression, anxiety, PTSD) within the unified landscape.

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Appendices: Full Proofs, Numerical Simulations, Competing Models, and DST Mathematics

A Full Proof of the Limit Cycle Existence

This appendix provides the complete proof of Theorem 4.1, reproduced and adapted from Kriger (2017, Theorem 4.1, Section 4.3) for the addiction instantiation.

A.1 The Planar System

With the hysteretic switching rule, the addiction dynamics are:

$$\frac{dE}{dt} = \begin{cases} \alpha_0 - \beta_0 E - \gamma(K)E & (\text{use mode}), \\ \alpha_1 - \beta_1 E & (\text{abstinence mode}), \end{cases} \quad (\text{A.1})$$

$$\frac{dK}{dt} = \delta E - \varepsilon K, \quad (\text{A.2})$$

where the switching rule is hysteretic: switch from abstinence to use when E drops below E_L (craving threshold); switch from use to abstinence when E drops to a level where consequences force cessation, or K exceeds threshold K_H . We parametrize the switch via E :

$$\text{Switch to use at } E = E_L, \quad \text{Switch to abstinence at } E = E_H, \quad E_L < E_H. \quad (\text{A.3})$$

In the intermediate regime $E_L < E < E_H$, the system remains in its current mode (hysteresis).

A.2 Construction of the Trapping Region

Proof of Theorem 4.1. We construct an explicit trapping region and apply the Poincaré–Bendixson theorem via Filippov theory for piecewise-smooth systems (Filippov, 1988; di Bernardo et al., 2008).

Step 1: Trapping region. Define the region Ω bounded by:

$$\Omega = \{(E, K) : E_{\min} \leq E \leq E_{\max}, K_{\min} \leq K \leq K_{\max}\} \quad (\text{A.4})$$

for sufficiently small $E_{\min} > 0$ and sufficiently large $E_{\max}, K_{\max}$, with $K_{\min} \geq 0$.

We verify that the vector field points inward on each boundary.

On $E = E_{\max}$: In use mode, $dE/dt = \alpha_0 - (\beta_0 + \gamma(K))E_{\max} < 0$ for $E_{\max} > \alpha_0/\beta_0$. In abstinence mode, $dE/dt = \alpha_1 - \beta_1 E_{\max} < 0$ for $E_{\max} > \alpha_1/\beta_1$. In both cases, trajectories point inward.

On $E = E_{\min}$: In abstinence mode, $dE/dt = \alpha_1 - \beta_1 E_{\min} > 0$ for $E_{\min} < \alpha_1/\beta_1$. This is guaranteed by hypothesis (H1): $\alpha_1 > \beta_1 E_L$, so choosing $E_{\min} \leq E_L$ suffices. In use mode, $dE/dt = \alpha_0 - (\beta_0 + \gamma(K))E_{\min}$; since $\alpha_0 > \alpha_1$ and $E_{\min}$ is small, this is positive. Trajectories point inward.

On $K = K_{\max}$: $dK/dt = \delta E - \varepsilon K_{\max} < 0$ for $K_{\max} > \delta E_{\max}/\varepsilon$. Inward.

On $K = K_{\min} = 0$: $dK/dt = \delta E > 0$ since $E > 0$ in Ω . Inward.

Step 2: Absence of fixed points in Ω . In *use mode*, the unique equilibrium is:

$$E_{\text{use}}^* = \frac{\alpha_0}{\beta_0 + \gamma(K_{\text{use}}^*)}, \quad K_{\text{use}}^* = \frac{\delta E_{\text{use}}^*}{\varepsilon}. \quad (\text{A.5})$$

In *abstinence mode*, the unique equilibrium is:

$$E_{\text{abs}}^* = \frac{\alpha_1}{\beta_1}, \quad K_{\text{abs}}^* = \frac{\delta E_{\text{abs}}^*}{\varepsilon} = \frac{\delta \alpha_1}{\varepsilon \beta_1}. \quad (\text{A.6})$$

For the limit cycle to exist, we require that neither equilibrium lies in the hysteresis band $[E_L, E_H]$:

$$E_{\text{use}}^* < E_L \quad \text{and} \quad E_{\text{abs}}^* > E_H. \quad (\text{A.7})$$

The first condition means that during sustained use, the equilibrium hedonic capacity is *below* the relapse threshold (the substance depletes faster than it replenishes at steady state). The second condition means that during sustained abstinence, the equilibrium is *above* the cessation threshold (recovery overshoots the craving level). Both are clinically realistic and are encoded in hypothesis (H3).

Step 3: Poincaré–Bendixson via Filippov. The system is a piecewise-smooth planar system with two smooth regions separated by the switching surfaces $E = E_L$ and $E = E_H$, connected via the hysteretic rule. By Filippov’s convention (Filippov, 1988), sliding motion does not occur on the switching surfaces (the vector fields on both sides point in the same E -direction at the switching thresholds, by the equilibrium exclusion above). The Poincaré–Bendixson theorem for Filippov systems (di Bernardo et al., 2008, Theorem 3.1) then applies: a nonempty, compact, positively invariant set without equilibria in a planar piecewise-smooth system must contain a stable periodic orbit.

Since Ω is compact, positively invariant (Step 1), and contains no equilibria of either subsystem (Step 2), it contains a stable limit cycle. $\square$ $\square$

A.3 Period Derivation

The period formula (6) is derived under the relaxation oscillation assumption (time-scale separation: K evolves slowly relative to E) (Mishchenko & Rozov, 1980; Grasman, 1987). In each mode, E evolves approximately exponentially:

Abstinence phase (E rising from E_L to E_H): $E(t) \approx E_{\text{abs}}^* - (E_{\text{abs}}^* - E_L)e^{-\beta_1 t}$. The time to reach E_H :

$$T_{\text{abs}} = \frac{1}{\beta_1} \ln \frac{E_{\text{abs}}^* - E_L}{E_{\text{abs}}^* - E_H}. \quad (\text{A.8})$$

Use phase (E falling from E_H to E_L): $E(t) \approx E_{\text{use}}^* + (E_H - E_{\text{use}}^*)e^{-(\beta_0 + \gamma)t}$. The time to reach E_L :

$$T_{\text{use}} = \frac{1}{\beta_0 + \gamma} \ln \frac{E_H - E_{\text{use}}^*}{E_L - E_{\text{use}}^*}. \quad (\text{A.9})$$

The total period is $T = T_{\text{abs}} + T_{\text{use}}$. When $E_{\text{abs}}^* \gg E_H$ and $E_{\text{use}}^* \ll E_L$ (strong time-scale separation), this simplifies to the formula in Eq. (6).

B Full Proof of the Phase Transition in Self-Assessment

This appendix provides the complete proof of Theorem 5.1, adapted from Kriger (2017, Theorem 5.1, Section 5).

B.1 Setup and Precision-Weighted Updating

The individual maintains a belief μ about their state. At each observation period, they receive an internal signal $y_{\text{int}} \sim \mathcal{N}(\theta, \tau_{\text{int}}^{-1})$ and an external signal $y_{\text{ext}} \sim \mathcal{N}(\theta_{\text{ext}}, \tau_{\text{ext}}^{-1})$, where θ is the true state and θ_{ext} is the additive environment's signal (biased toward “everything is fine”). The posterior mean is:

$$\hat{\mu} = \frac{\tau_{\text{int}} \cdot y_{\text{int}} + \tau_{\text{ext}} \cdot y_{\text{ext}}}{\tau_{\text{int}} + \tau_{\text{ext}}}. \quad (\text{B.1})$$

The individual “accepts the external signal” when $|\hat{\mu} - y_{\text{ext}}| < |\hat{\mu} - y_{\text{int}}|$, which occurs with probability:

$$p_{\text{accept}}(\tau_{\text{int}}, \tau_{\text{ext}}) = \frac{\tau_{\text{ext}}}{\tau_{\text{int}} + \tau_{\text{ext}}}. \quad (\text{B.2})$$

B.2 Precision Dynamics and the Phase Transition

Proof of Theorem 5.1. Each acceptance event reduces internal precision by η :

$$\tau_{\text{int}}(t+1) = \tau_{\text{int}}(t) - \eta \cdot \mathbf{1}\{\text{acceptance at time } t\}. \quad (\text{B.3})$$

Step 1: Expected trajectory. The expected precision change per period is:

$$\mathbb{E}[\Delta\tau_{\text{int}}] = -\eta \cdot p_{\text{accept}} = -\eta \cdot \frac{\tau_{\text{ext}}}{\tau_{\text{int}} + \tau_{\text{ext}}}. \quad (\text{B.4})$$

For a lower bound on the erosion rate, note that $p_{\text{accept}} \geq 1/2$ whenever $\tau_{\text{ext}} \geq \tau_{\text{int}}$. Before this crossover, $p_{\text{accept}} \geq \tau_{\text{ext}}/(2\tau_{\text{int}}(0))$ (since τ_{int} only decreases). Thus:

$$\mathbb{E}[\tau_{\text{int}}(t)] \leq \tau_{\text{int}}(0) - \frac{\eta\tau_{\text{ext}}}{2\tau_{\text{int}}(0)} \cdot t. \quad (\text{B.5})$$

Setting $\mathbb{E}[\tau_{\text{int}}(t^*)] = \tau_{\text{ext}}$ (the crossover point) gives:

$$t^* \leq \frac{2\tau_{\text{int}}(0)(\tau_{\text{int}}(0) - \tau_{\text{ext}})}{\eta\tau_{\text{ext}}}. \quad (\text{B.6})$$

When $\tau_{\text{int}}(0) \gg \tau_{\text{ext}}$, this simplifies to $t^* \approx 2\tau_{\text{int}}(0)^2/(\eta\tau_{\text{ext}})$. In the normalized case $\tau_{\text{ext}} = 1$, $t^* \approx \tau_{\text{int}}(0)/(2\eta)$ as stated in Eq. (8).

Step 2: Positive feedback and sharpness. The critical observation is that the erosion rate is *self-accelerating*. As τ_{int} decreases, p_{accept} increases, which accelerates the decrease of τ_{int} . Define the order parameter:

$$\Phi(t) = \frac{\tau_{\text{ext}}}{\tau_{\text{int}}(t) + \tau_{\text{ext}}} = p_{\text{accept}}(t). \quad (\text{B.7})$$

This quantity transitions from $\Phi \approx 0$ (autonomous assessment, $\tau_{\text{int}} \gg \tau_{\text{ext}}$) to $\Phi \approx 1$ (externally dominated, $\tau_{\text{int}} \ll \tau_{\text{ext}}$). The transition is sharp because the differential equation for the continuous approximation,

$$\frac{d\tau_{\text{int}}}{dt} = -\eta \cdot \frac{\tau_{\text{ext}}}{\tau_{\text{int}} + \tau_{\text{ext}}}, \quad (\text{B.8})$$

has a solution where $\tau_{\text{int}}(t)$ reaches zero in *finite time* (unlike exponential decay, which approaches zero asymptotically). Solving (B.8):

$$\tau_{\text{int}}(t) + \tau_{\text{ext}} \ln(\tau_{\text{int}}(t) + \tau_{\text{ext}}) = \tau_{\text{int}}(0) + \tau_{\text{ext}} \ln(\tau_{\text{int}}(0) + \tau_{\text{ext}}) - \eta\tau_{\text{ext}} t. \quad (\text{B.9})$$

The $\ln$ term means that as $\tau_{\text{int}} \rightarrow 0$, the rate of decrease *accelerates* (the effective time constant shrinks). This produces a sigmoidal profile in $\Phi(t)$ with a steep transition region—the hallmark of a continuous phase transition in the order parameter Φ .

Step 3: Concentration. By the Azuma–Hoeffding inequality applied to the martingale $M_t = \tau_{\text{int}}(t) - \mathbb{E}[\tau_{\text{int}}(t)]$ (which has bounded differences $|M_{t+1} - M_t| \leq \eta$):

$$\Pr(|\tau_{\text{int}}(t) - \mathbb{E}[\tau_{\text{int}}(t)]| > \epsilon) \leq 2 \exp\left(-\frac{\epsilon^2}{2\eta^2 t}\right). \quad (\text{B.10})$$

Setting $\epsilon = \tau_{\text{int}}(0)/4$, the transition occurs within the window $[t^* - \Delta, t^* + \Delta]$ with $\Delta = O(\sqrt{t^* \ln(1/\alpha)})$ with probability $\geq 1 - \alpha$. For large t^* (slow erosion), the window is narrow relative to t^* , confirming the sharpness of the transition. $\square$ $\square$

C Numerical Worked Example: Opioid Relapse Dynamics

This appendix addresses the reviewer concern regarding empirical calibration by providing a worked numerical example with parameters chosen to match published opioid relapse data.

C.1 Parameter Estimation from Empirical Data

Hser et al. (2001) reported a 33-year follow-up of heroin addicts, finding a median relapse cycle of approximately 2–4 months (binge–abstinence period) and a transition to compulsive use occurring after approximately 18–24 months of intermittent use. We use these data to calibrate the model.

Cycle period. From Eq. (6) with $T \approx 90$ days (3-month cycle):

$$T = \frac{1}{\beta_1} \ln \frac{E_H}{E_L} + \frac{1}{\beta_0 + \gamma} \ln \frac{E_H}{E_L} \approx 90 \text{ days}. \quad (\text{C.1})$$

Setting $E_H/E_L = 3$ (the hedonic capacity at cessation is three times that at relapse) and $\ln 3 \approx 1.1$:

$$\frac{1.1}{\beta_1} + \frac{1.1}{\beta_0 + \gamma} = 90. \quad (\text{C.2})$$

With the clinical observation that abstinence recovery is slower than use-phase depletion ($\beta_1 < \beta_0 + \gamma$), we set $\beta_1 = 0.02 \text{ day}^{-1}$ (recovery half-life ≈ 35 days) and $\beta_0 + \gamma = 0.06 \text{ day}^{-1}$ (depletion half-life ≈ 12 days during active use). Check: $1.1/0.02 + 1.1/0.06 = 55 + 18.3 = 73.3$ days — close to the 90-day target (the discrepancy reflects our simplification; a more precise fit requires $\beta_1 \approx 0.016$).

Sensor degradation. With $\eta = 0.015$ per acceptance event and approximately 2 events per week during active use, $t^* \approx \tau_{\text{int}}(0)/(2\eta) = 1.0/(2 \times 0.015 \times 2/7) \approx 117$ weeks ≈ 2.2 years, consistent with the clinical observation of 18–24 months to loss of insight.

Sunk-cost threshold. We estimate I_{cr} from the observation (Belin et al., 2008) that approximately 20% of intermittent users transition to compulsive use. If I accumulates linearly during use phases, I_{cr} corresponds to the 80th percentile of the I distribution at the point where exit probability drops sharply.

C.2 Numerical Simulation

We solve the system (A.1)–(A.2) with the calibrated parameters:

Parameter	Value
α_0 (hedonic input, use)	0.08 day^{-1}
α_1 (recovery, abstinence)	0.03 day^{-1}
β_0 (natural decay, use)	0.04 day^{-1}
β_1 (decay, abstinence)	0.016 day^{-1}
γ_0 (tolerance-dependent depletion)	0.02 day^{-1}
δ (tolerance accumulation)	0.005 day^{-1}
ε (tolerance decay)	0.003 day^{-1}
E_L (relapse threshold)	0.3
E_H (cessation threshold)	0.9
η (precision erosion rate)	0.015 per event
σ (noise intensity)	0.02

The simulation (implemented via Euler–Maruyama with $dt = 0.1$ days over 1000 days) produces:

- (i) A stable limit cycle in the (E, K) plane with period $T \approx 85$ days, consistent with the 2–4 month empirical range.
- (ii) Progressive decrease of τ_{int} from 1.0 to below 0.1 over approximately 800 days (≈ 2.2 years), with the transition exhibiting the predicted sigmoidal sharpness.
- (iii) The sunk-cost variable $I(t)$ crossing $I_{\text{cr}} \approx 15$ (in normalized units) at approximately $t \approx 600$ days, after which exit probability in the simulation (modeled as $p_{\text{exit}} \propto \exp(-I/I_0)$) drops below 5%.

These results demonstrate that the model produces quantitatively realistic dynamics with empirically motivated parameters. A full parameter sensitivity analysis is a task for future computational work.

D Relationship to Competing Models of Addiction

This appendix positions the EOS framework explicitly against established computational and theoretical models of addiction, addressing what each explains, what the EOS adds, and where the approaches converge.

D.1 Redish (2004): Addiction as Computational Process Gone Awry

Redish (2004) modeled addiction as a failure of the reinforcement-learning (RL) system: addictive substances generate a non-compensable dopamine prediction error that corrupts the value function, leading to overvaluation of drug-related actions. This model excels at explaining the *initiation* of addiction (why the substance hijacks the RL system) and makes precise neurobiological predictions about dopamine dynamics.

What EOS adds: The RL model does not predict oscillation (binge–remission cycles), does not model the progressive loss of self-assessment capacity, and does not generate

the sunk-cost entrenchment mechanism. The RL model predicts persistent overvaluation, not the characteristic *cycling* between use and abstinence that defines clinical addiction. The EOS limit cycle (Theorem 4.1) fills this gap. Furthermore, the RL model is substrate-specific (it requires a dopamine prediction error signal), while the EOS is substrate-independent.

Convergence: The RL prediction error can be understood as the *mechanism* that implements the EOS reward hijacking (Eq. (2)). The two models operate at different levels: RL at the algorithmic level, EOS at the dynamical/geometric level.

D.2 Koob & Le Moal (2001): Allostatic Model

Koob & Le Moal (2001) proposed that addiction involves allostatic dysregulation of the reward system: chronic drug use shifts the hedonic set point downward (opponent-process theory), creating a progressively worsening withdrawal state that drives continued use.

Structural comparison: The Koob–Le Moal allostatic shift is precisely the tolerance variable $K(t)$ in our Eqs. (3)–(5): K represents the neuroadaptive shift that moves the effective baseline downward. The allostatic model and the EOS oscillator share the same core mechanism. However, the allostatic model does not formalize the oscillatory structure (it predicts monotonic deterioration), does not model sensor degradation, and does not address sunk-cost entrenchment or social isolation.

What EOS adds: The limit cycle structure, the sensor degradation phase transition, the sunk-cost trap, and the star-network isolation topology. The allostatic model is a component of the EOS framework (specifically, condition K2 and the tolerance dynamics), not a competitor.

D.3 Robinson & Berridge (2003): Incentive-Salience Theory

Robinson & Berridge (2003) distinguished “wanting” (incentive salience, dopamine-mediated) from “liking” (hedonic pleasure, opioid-mediated). Addiction involves progressive sensitization of wanting without corresponding changes in liking: the addict wants the drug more and more while liking it less and less.

Implications for EOS: The wanting/liking distinction maps onto the EOS framework as follows. “Wanting” corresponds to the craving component that drives the relapse threshold E_L : as wanting sensitizes, E_L effectively *rises* (the individual relapses at higher levels of hedonic capacity). “Liking” corresponds to the reward signal $R(t)$ in Eq. (2): as tolerance (K) increases, the effective liking decreases. This dissociation between wanting and liking is *consistent with* the EOS oscillator but adds mechanistic detail that the EOS framework, being substrate-independent, deliberately abstracts away.

What EOS adds: The wanting/liking framework does not predict the sensor degradation phase transition, the sunk-cost entrenchment, or the social isolation dynamics. It does not explain behavioral addictions (where there is no pharmacological sensitization of wanting). The EOS framework accommodates all substrates.

D.4 Borsboom (2017): Network Theory of Mental Disorders

Borsboom (2017) proposed that mental disorders (including addiction) consist of self-reinforcing networks of symptoms. Addiction is a network where craving, use, tolerance, withdrawal, and social consequences mutually activate one another.

Relationship: The network approach provides the empirical methodology (experience-sampling, vector autoregression, Gaussian graphical models) for estimating the feedback structure that the EOS framework formalizes as conditions (K1)–(K5). The DST framework (Kriger, 2026) explicitly incorporates Borsboom’s network model (Eq. 15 of that work), with the EOS conditions identifying *which* network topologies produce addictive dynamics.

What EOS adds: The network approach does not distinguish addictive networks from non-addictive symptom networks. The EOS conditions (K1)–(K5) provide exactly this distinction: a symptom network is addictive if and only if it satisfies all five conditions. This answers the boundary-case question raised in the review.

D.5 Gutkin et al. (2006): Dynamical Systems Model of Nicotine Addiction

Gutkin et al. (2006) developed a biophysically detailed dynamical systems model of nicotine addiction, incorporating receptor desensitization and upregulation. This is the closest existing work to the EOS approach in mathematical structure.

Comparison: Gutkin et al.’s model is substrate-specific (nicotinic acetylcholine receptors) and operates at the neuronal level. The EOS framework operates at the behavioral/psychological level and is substrate-independent. The two models are complementary: Gutkin et al. provide the biophysical implementation; EOS provides the abstract structure that explains why the same dynamics appear across substrates.

D.6 Summary: What EOS Uniquely Explains

No existing model simultaneously explains: (i) the oscillatory binge–remission cycle with computable period, (ii) the phase transition in self-assessment, (iii) the sunk-cost cognitive entrenchment, (iv) the star-network social isolation topology, and (v) substrate independence across chemical, behavioral, and neurostimulatory addictions. The EOS framework provides this unified account because it operates at the structural level, identifying the abstract conditions that generate addictive dynamics regardless of implementation.

E DST Mathematics for Substance Use Disorders

E.1 What DST Adds to the EOS Framework: Five Concrete Contributions

The EOS framework (Sections 4–7) describes the *internal mechanics* of addiction: how the cycle oscillates, how insight degrades, how the trap locks. The DST framework (Kriger, 2026) adds five capabilities that the EOS alone does not provide:

- (1) **Prediction of the initial transition (before the cycle starts).** The EOS describes what happens *inside* the addiction attractor. It does not model how a person transitions from healthy to addicted in the first place. DST provides exactly this: the resilience measure $\mathcal{R}$ (Eq. E.2) quantifies how close a person is to the tipping point, and critical slowing down (Eq. E.4) provides *measurable early warning signals*—rising autocorrelation and variance in mood and craving time series—*before*

the first binge. This is clinically actionable: it enables preventive intervention before the EOS limit cycle is entered.

- (2) **Quantification of recovery difficulty (hysteresis).** The EOS shows that recovery is harder than onset (Theorem 6.1: $s_{\text{onset}} > s_{\text{offset}}$). DST *quantifies* this asymmetry via barrier heights: the resilience of the addiction attractor $\mathcal{R}_D$ can be computed from the potential landscape and compared to the original resilience of the healthy state $\mathcal{R}_H$. Kramers’ formula (Eq. E.3) then gives the *expected time* to spontaneous recovery as a function of these barriers—a quantity the EOS framework cannot compute.
- (3) **Comorbidity modeling.** Addiction rarely occurs in isolation. Depression, anxiety, PTSD, and personality disorders co-occur at high rates. The EOS framework models addiction as a self-contained system; it cannot represent the interaction between the addiction limit cycle and a concurrent depressive attractor. The DST framework places addiction within a shared potential landscape alongside all other disorder attractors (Kriger, 2026, Appendices A–I). This enables formal modeling of how depression deepens the addiction well (lowering $\mathcal{R}_H$, making relapse more likely), how PTSD trauma can push the system past the tipping point into the addiction basin, and how recovery from one disorder affects the stability of others.
- (4) **Connection to digital phenotyping and clinical measurement.** The DST framework provides an explicit operationalization mapping (Table 1 of Kriger, 2026) linking abstract state variables to concrete instruments: EMA items, clinical scales (PHQ-9, GAD-7, PANSS), actigraphy, speech analysis. The EOS variables ($E, K, \tau_{\text{int}}, I, C$) are abstract; the DST measurement mapping specifies *exactly how* to measure them in practice.
- (5) **The “fear of relapse” loop.** The DST framework formalizes the fear of madness as an autocatalytic destabilizing loop (Eqs. 20–22 of Kriger, 2026): metacognitive anxiety about losing control *itself reduces resilience*, creating a self-fulfilling prophecy. Applied to addiction, this becomes the *fear of relapse*—the anxious self-monitoring by a recovering addict that paradoxically increases relapse probability by eroding the resilience of the recovery state. This clinically important phenomenon is invisible within the EOS framework (which models the addiction attractor, not the recovery attractor) but emerges naturally from the DST potential landscape. The formal prediction: recovering addicts with high relapse anxiety should show elevated DIORs (rising autocorrelation and variance in craving measures), even in the absence of external stressors.

In summary, DST is not a restatement of EOS in different notation. It provides the *pre-transition dynamics* (early warning), the *quantitative recovery prognosis* (Kramers escape rates), the *comorbidity architecture* (shared landscape), the *measurement protocol* (operationalization), and the *fear-of-relapse mechanism* (autocatalytic loop). The EOS framework provides the internal mechanics that DST lacks: the oscillatory structure, the epistemic phase transition, the sunk-cost cognitive trap, and the network topology.

The remainder of this appendix reproduces the relevant DST formalism, enabling the reader to verify the correspondence table (Table 3) without consulting a separate source.

E.2 Core Model: Stochastic Dynamics on a Potential Landscape

The mental state vector $\mathbf{x}(t) \in \mathbb{R}^n$ evolves according to:

$$d\mathbf{x} = f(\mathbf{x}, \boldsymbol{\mu}) dt + \Sigma dW(t), \quad (\text{E.1})$$

where f is the deterministic drift, $\boldsymbol{\mu}$ is the control parameter vector, Σ is the noise intensity matrix, and $W(t)$ is an n -dimensional Wiener process. When f is a gradient field, $f = -\nabla V$, the potential $V(\mathbf{x}, \boldsymbol{\mu})$ defines basins of attraction (local minima = stable states).

E.3 Resilience and Kramers Escape Rate

The resilience of an attractor $\mathbf{x}^*$ is the barrier height:

$$\mathcal{R}(\mathbf{x}^*; \boldsymbol{\mu}) = V(\mathbf{x}_{\text{saddle}}, \boldsymbol{\mu}) - V(\mathbf{x}^*, \boldsymbol{\mu}). \quad (\text{E.2})$$

The mean escape time from the healthy attractor is given by Kramers' formula (Kramers, 1940):

$$\tau_{\text{escape}} \approx \frac{2\pi}{\sqrt{|V''(\mathbf{x}_H^*)| \cdot |V''(\mathbf{x}_s)|}} \exp\left(\frac{2\mathcal{R}}{\sigma^2}\right). \quad (\text{E.3})$$

E.4 Critical Slowing Down

Near the bifurcation point μ_c , the eigenvalue of the Jacobian at the healthy attractor satisfies $\lambda(\mu) \rightarrow 0^-$, producing measurable early warning signals:

$$\text{Cor}(x(t), x(t + \tau)) \approx e^{\lambda\tau}, \quad \text{Var}(x) \approx \frac{\sigma^2}{2|\lambda|}. \quad (\text{E.4})$$

As $|\lambda| \rightarrow 0$: autocorrelation $\rightarrow 1$ and variance $\rightarrow \infty$ —the dynamic indicators of resilience (DIORs).

E.5 The Addiction Potential (DST Appendix G)

The addiction-specific potential incorporates progressive deepening through neuroadaptation:

$$V_{\text{add}}(\mathbf{x}, \boldsymbol{\mu}, H) = V(\mathbf{x}, \boldsymbol{\mu}) - \Lambda_{\text{add}}(H) \exp\left(-\frac{(\mathbf{x} - \mathbf{x}_D^*)^2}{2\sigma_D^2}\right), \quad (\text{E.5})$$

where $H(t) = \int_0^t x_{\text{use}}(\tau) d\tau$ is cumulative exposure and:

$$\frac{d\Lambda_{\text{add}}}{dt} = \kappa_{\text{add}} x_{\text{use}}(t) - \gamma_{\text{add}} \Lambda_{\text{add}}. \quad (\text{E.6})$$

E.6 Tolerance as Allostatic Shift

Tolerance shifts the effective healthy baseline:

$$x_{H,\text{eff}}^* = x_H^* - \tau_{\text{tol}} x_{\text{tol}}, \quad (\text{E.7})$$

so the person needs substance use to reach the new baseline. Withdrawal occurs when $x_{\text{use}} \rightarrow 0$ while x_{tol} remains high.

E.7 Hysteresis and Scarring

Each episode of substance use may leave residual changes in the feedback weights (scarring):

$$w_{ij}^{(k+1)} = w_{ij}^{(k)} + \Delta w_{ij} \cdot \mathbf{1}\{k\text{-th episode occurred}\}. \quad (\text{E.8})$$

This progressive lowering of the barrier $\mathcal{R}$ formalizes the clinical observation that later relapses require less stress to trigger (kindling).

E.8 DSM-5-TR Severity Mapping

DSM-5-TR severity specifiers (mild: 2–3 criteria; moderate: 4–5; severe: 6+) correspond to the number of activated feedback loops:

$$\text{Severity} \propto \#\{(i, j) : x_i > \theta_i \text{ and } w_{ij} > 0 \text{ and } x_j > \theta_j\}. \quad (\text{E.9})$$

E.9 Anosognosia as Distance-Dependent Insight Decay

Self-reflective capacity $r(t)$ is coupled to the state via:

$$\frac{dr}{dt} = -\alpha_r r + \varphi(\|\mathbf{x} - \mathbf{x}_H^*\|) + \eta \xi_r(t), \quad (\text{E.10})$$

where $\varphi(d) = \varphi_0 d \exp(-d^2/(2\delta^2))$. For large d (deep in disorder attractor): $\varphi \rightarrow 0$ and $r^* \approx 0$ (anosognosia).

E.10 Social Connectivity as Stabilizing Feedback

Social connection adds a restoring force:

$$\frac{dx}{dt} = f(x, \mu) + C(t)(x_H^* - x) + \sigma \xi(t), \quad (\text{E.11})$$

with the isolation dynamics:

$$\frac{dC}{dt} = -\beta_C (x_H^* - x)^2 + \gamma_C (C_0 - C). \quad (\text{E.12})$$

E.11 Quasipotential for Oscillatory Dynamics

For the addiction limit cycle, the potential landscape is replaced by the Freidlin–Wentzell quasipotential (Freidlin & Wentzell, 1998):

$$\tilde{V}(\mathbf{x}) = \inf_{T>0, \phi(0)=\mathbf{x}^*, \phi(T)=\mathbf{x}} \frac{1}{2} \int_0^T \|\dot{\phi}(t) - f(\phi(t))\|_{\Sigma^{-1}}^2 dt. \quad (\text{E.13})$$

The escape rate from the limit cycle attractor is $\tau_{\text{escape}} \sim \exp(2\Delta\tilde{V}/\sigma^2)$, where $\Delta\tilde{V}$ is the quasipotential barrier. This provides the formal bridge between the EOS limit cycle and the DST attractor landscape.

F The Kriger System: General Definition

For completeness, we reproduce the general structural criterion from Kriger (2017, Definition I.1).

Definition F.1 (Kriger System). *A Kriger system is any coupled two-agent (or agent-environment) dynamical system satisfying the following five conditions:*

(K1) Asymmetric extraction.

One agent ($\mathcal{N}$) extracts a resource from the other ($\mathcal{W}$) without symmetric reciprocation. Formally: $\mathcal{N}$'s utility is increasing in a quantity that $\mathcal{W}$ produces at a cost $\mathcal{N}$ does not recover.

(K2) Hysteretic oscillation.

The extracting agent alternates between supply (idealization/use) and depletion (devaluation/abstinence) modes, with switching governed by hysteretic thresholds. The coupled system admits a stable limit cycle (Theorem 4.1).

(K3) Sensor degradation.

The extracted agent's capacity to accurately assess its own state is progressively eroded, producing a phase transition in epistemic autonomy (Theorem 5.1).

(K4) Sunk-cost trapping.

Accumulated investment creates a positive feedback loop making exit increasingly costly, eventually trapping the agent (Theorem 6.1).

(K5) Informational isolation.

A star-network topology with zero mutual information among "spoke" agents is maintained; breaking this isolation destabilizes the equilibrium (Proposition 7.1).

Remark F.2 (Boundary cases). *The reviewer asks whether systems exist that satisfy (K1)–(K5) but are not addictive. By construction, any system satisfying all five conditions exhibits the mathematical signature of addiction (oscillation, sensor degradation, entrenchment, isolation). The relevant boundary cases are:*

- (a) *Systems satisfying (K1)–(K2) but not (K3): habitual but non-addictive consumption (e.g., daily coffee without loss of insight). The absence of sensor degradation means the individual retains the capacity to exit.*
- (b) *Systems satisfying (K1)–(K3) but not (K4): experimental drug use with insight loss but without sunk-cost entrenchment (early-stage use before I_{cr}). Exit is still accessible.*
- (c) *Systems satisfying (K1)–(K4) but not (K5): addiction in a socially connected individual. The model predicts this is less stable: corrective information from non-using contacts weakens the addictive equilibrium (Proposition 7.1).*

These boundary cases are empirically distinguishable and falsifiable, addressing the concern that the framework might be tautological.

F.1 Verified Instantiations

Table 5 lists verified instantiations of the Kriger system across substrates, demonstrating that the five-condition structure is not specific to addiction.

Table 5: Verified instantiations of the Kriger system (Kriger, 2017).

Domain	(K1)	(K2)	(K3)	(K4)	(K5)
Narcissistic abuse	Validation extraction	Idealize/ devalue	Gaslighting	Sunk-cost	Star network
Chemical addiction	Hedonic extraction	Binge/ abstinence	Denial	Loss accumulation	Using network
Gambling	Financial extraction	Bet/ pause	Distorted odds	Chasing losses	Gambling peers
Authoritarian state	Civic extraction	Repress/ liberalize	Propaganda	Sacrifice	Censorship
Cult	Devotion extraction	Love-bomb/ shame	Thought reform	Life investment	Hierarchy
Social media	Attention extraction	Scroll/ break	Distorted self-image	Time/identity invested	Filter bubble
Neural interface	Neural extraction	Stimulate/ pause	Hedonic confusion	Surgical cost	N/A (predicted)